

The Road Less Traveled: Strategy Distinctiveness and Hedge Fund Performance

Zheng Sun

Paul Merage School of Business, University of California Irvine, CA 92697-3125

Phone: (949) 824-6907

Email: zsun@merage.uci.edu

Ashley Wang

Board of Governors of the Federal Reserve System at Washington DC

Email: ashley.wang@frb.gov

Lu Zheng

Paul Merage School of Business, University of California Irvine, CA 92697-3125

Phone: (949) 824-8365

Email: luzheng@uci.edu

Abstract

Basic economic principle suggests that a well-known trading strategy offers little economic profit. In this paper, we plan to investigate whether skilled hedge fund managers are more likely to pursue unique investment strategies that will result in superior performance. We propose a measure of the distinctiveness of a fund's investment strategy based on historical fund return data. Specifically, we plan to examine the extent to which a fund's returns differ from those of its peer funds. We term the measure "Strategy Distinctiveness Index" (*SDI*). The higher the *SDI*, the more distinctive is a fund's strategy. We hypothesize that the more distinctive a fund's investment strategy, the more innovative and skillful the fund manager, and the better future performance. We will conduct empirical tests to probe whether funds with higher *SDI* indeed offer better subsequent risk adjusted performance. The empirical findings will contribute to the growing literature on the cross-sectional determinants and predictors of hedge fund performance.

I. Research Objective

Investors pay high fees for hedge fund performance¹. Basic economic principle suggests that only unique investment ideas would generate superior performance because any potential abnormal return of a well-known and heavily traded strategy is likely to be competed away. Therefore, identifying those fund managers possessed of unique investment ideas is crucial for hedge fund investors. In this paper, we make an initial attempt to estimate the uniqueness and distinctiveness of a fund's investment strategy. Further, we plan to examine whether a distinctive investment strategy is indicative of greater managerial talents, and hence, superior fund performance. The empirical findings will contribute to the growing literature on the cross-sectional determinants and predictors of hedge fund performance.²

As discussed in Stein's AFA presidential address (2009), the "zero-profit" condition for a competitive economy suggests that "enough money chasing a given pattern in returns will necessarily eliminate that pattern." Competitions among professional money managers have intensified over the past 20 years due to the vast growth of the hedge fund industry.³ The fierce competition quickly diminishes a strategy's economic profit as it becomes well-known. Therefore, fund managers need to exploit new investment ideas to generate superior performance. Consistent with the "zero-profit" intuition, recent papers show that profitable ideas are likely to be guarded and stay localized, while only less valuable investment ideas tend to spread out.⁴ Anecdotal evidence also indicates that hedge funds protect their

¹ Research on hedge fund performance in general suggests that hedge funds deliver positive excess returns, while the evidence on performance persistence has been rather mixed. See for example, Ackermann, McEnally, and Ravenscraft (1999) Agarwal and Naik (2000 and 2004), Brown, Goetzmann, and Ibbotson (1999), Brown and Goetzmann (2003), Brown, Goetzmann, Liang, and Schwarz (2007), Fung and Hsieh (1997, 2000, 2001, 2002), Goetzmann, Ingersoll, and Ross (2003), Griffin and Xu (2007), Ibbotson and Chen (2006), Jagannathan, Malakhov, and Novikov (2006), Kosowski, Naik, and Teo (2007) and Liang (1999, 2000).

² Recent papers include Agarwal, Daniel, and Naik (2007), Aggarwal and Jorion (2009), Aragon (2007), Liang and Park (2008), Li, Zhang, and Zhao (2007), Malkiel and Saha(2005), Titman and Tiu (2008).

³ The total AUM of funds reporting to TASS has grown from \$50 billion in 1994 to \$1.2 trillion in 2008.

⁴ Please see Stein (2008) and Gray (2009) for theoretical and empirical evidences.

investment ideas by all means.⁵ Detecting new great ideas, however, is costly and requires skills. Thus skilled managers are more likely to generate and pursue unique investment strategies that will result in superior performance, while less skilled managers are more likely to trade on known strategies. Following this hypothesis, we should observe a positive relation between distinctiveness in fund strategy and fund performance.

Moreover, hedge fund managers pursuing distinctive strategies may be less subject to negative externalities due to the “crowded-trade” effect and the leverage effect, both elaborated in Stein (2009). The “crowded-trade” effect occurs when an arbitrageur faces additional price uncertainty due to the inability to know how many others are using the same model and taking the same position as her. In addition, when traders follow the same set of signals and buy the same assets using leverage, they may incur significant losses in asset value if one of their peers holding similar portfolios is hit with a negative shock and is forced to liquidate her assets in a fire sale (the leverage effect). These negative externalities, especially imposed on funds trading similarly to others, increase the risk of the trading strategies and may make ex-ante profitable investment strategies lose money ex-post.

Nevertheless, there also exists counteracting mechanisms that may dampen or even reverse the positive association between the distinctiveness of hedge fund strategies and future performance. First, unskilled managers may take excessive idiosyncratic risk, due to a potential conflict of interest between fund managers and investors. The option-like feature of the compensation contract may create incentives for hedge fund managers to make idiosyncratic bets to increase the chance of extreme performance.⁶ Funds pursuing such gaming strategy would appear to be distinctive from the peers yet with no superior performance. The second argument is rooted in the limits of arbitrage. For example, when individual

⁵ For example, hedge fund industry strongly opposed to the 2004 SEC disclosure requirement. In 2006, the US Court vacated the new rule. In addition, hedge funds frequently file trade-secrets lawsuits against former employees for using their proprietary trading strategies.

arbitrageurs face capital constraints, coordination and synchronization among multiple arbitrageurs may be necessary to successfully correct mispricing⁷. Therefore, hedge funds that do not coordinate with their peers may suffer temporary yet significant losses, especially in circumstances when noise trader risk is high, such as bubble periods. In these cases, we would expect a negative relation or no relation between distinctiveness in fund strategy and fund performance.

Ultimately, whether and how the distinctiveness of hedge fund strategy is associated with future fund risk adjusted performance is an empirical question, and the current literature is silent on this issue. This project intends to tackle this task.

II. Previous Research in the Area

Despite the importance of distinguishing skilled hedge fund managers from unskillful ones, research on the cross-sectional determinants of hedge fund returns has been rather limited until several recent papers started linking hedge fund performance to various fund and managerial attributes. Aragon (2007) and Liang and Park (2008) find that funds with more stringent share restriction clauses offer higher returns. Agarwal, Daniel, and Naik (2007) show that funds offering greater managerial incentives and discretion display superior performance. Aggarwal and Jorion (2009) document strong outperformance by emerging hedge fund managers, especially during the first 2 to 3 years of fund existence. Li, Zhang, and Zhao (2007) find that educational background and working experience of managers are related to hedge fund performance. Titman and Tiu (2008) argue that skilled managers choose to hedge. They show that funds with lower *R*-squares with respect to systematic risk factors subsequently outperform.

The existing literature examining the effect of the innovativeness of managerial talents and distinctiveness of fund strategy on fund performance has primarily focused on the mutual fund sector.

⁶ Please see Goetzmann, Ingersoll, and Ross (2003).

Kacperczyk, Sialm, and Zheng (2005) argue that mutual fund managers may decide to deviate from a well-diversified portfolio and concentrate their holdings in industries in which they have informational advantages. Their results confirm that more concentrated funds perform better, after controlling for risk and style differences. In a related paper, Cremers and Petajisto (2007) propose a measure of Active Share for individual mutual funds to capture the share of portfolio holdings that differ from the benchmark index. They find that funds with the highest Active Share values significantly outperform their benchmark, both before and after expenses.

Several papers propose to evaluate mutual fund performance by their use of public information, similarity of holdings to star funds and the effect of unobserved actions. Kacperczyk and Seru (2007) construct a Reliance on Public Information (*RPI*) measure to capture the responsiveness of a mutual fund manager's portfolio allocations to changes in public information, and they find a strong inverse relation between the *RPI* and future fund performance. Cohen, Coval, and Pastor (2005) propose to judge a fund manager's skill by how similar her portfolio holdings are to those of managers with superior performance records. They demonstrate empirically that their measures are useful in forecasting manager performance. Kacperczyk, Sialm, and Zheng (2007) propose a return gap measure, the difference between the reported fund returns and the return of a portfolio that invests in the previously disclosed holding, to capture the unobserved actions taken by mutual fund managers. They find that the return gap significantly predicts future fund performance.

In this paper, we try to estimate the distinctiveness of a hedge fund strategy, a previously unstudied aspect of managerial quality. This task is especially important but challenging for hedge funds. First, hedge fund managers conduct their trading operations amid great secrecy, offering little disclosure in order to protect their investment ideas. Second, the rapid growth of the hedge fund industry has resulted in a wide range of strategies and a huge number of funds run by managers with diverse investment

⁷ This is referred to as synchronization risk in Abreu and Brunnermeier(2002).

backgrounds and qualifications. Our paper attempts to assess uniqueness of hedge fund strategies by analyzing the limited fund information in the public domain.

III. Research Design and Methods

This project intends to investigate whether a more distinctive investment strategy reflects more innovative and skillful managerial talents, hence predicting superior future performance. To construct the distinctiveness measure and fund performance, we will assemble a panel of hedge fund data from Lipper TASS database for the period of January of 1994 to December of 2009, including monthly hedge fund returns, as well as fund characteristics.

1. Quantifying Hedge Fund Distinctiveness

If a manager is skillful, she is likely to engage in an innovative and unique trading strategy, thereby delivering performance that co-moves less with the overall performance of the hedge fund sector, or the performance of the specific style to which her fund belongs. This suggests an intuitive measure to capture the distinctiveness of a fund strategy: one minus the sample correlation of a fund's return (r_{it}) with the average return of all funds belonging to the same style (μ_{It}):

$$\begin{aligned} SDI_i &= 1 - \text{corr}(r_i, \mu_I) \\ &= 1 - \frac{\sum_{t=1}^{24} (r_{it} - \bar{r}_i)(\mu_{It} - \bar{\mu}_I)}{\sqrt{\sum_{t=1}^{24} (r_{it} - \bar{r}_i)^2 \sum_{t=1}^{24} (\mu_{It} - \bar{\mu}_I)^2}} \quad (1) \end{aligned}$$

where $\mu_{It} = \frac{\sum_{j \in I} r_{jt}}{\text{count}(j \in I)}$. The SDI ranges between 0 and 2 in theory. Graphically, SDI can be viewed as a “distance” measure: the higher the SDI , the farther a fund is from its cluster and the more distinctive must be the fund's strategy.

To gauge how distinctive a fund's strategy is from its cohort, we need to define hedge fund styles appropriately. Although TASS offers a classification scheme of 10 styles based on survey and voluntary

reporting of hedge fund managers, this classification has a number of limitations such as strategic misclassification and time-invariant styles.

To address these issues, this paper plans to endogenously and dynamically classify fund styles by clustering historic returns. At the beginning of each quarter, we will group them into K clusters (ie. K styles) based on the correlation of fund returns. The clustering procedure will be similar to the method in Brown and Goetzmann (1997, 2003). The goal of the procedure is to find a locally optimized partition among funds, so that it minimizes the sum of the distance of all funds to the corresponding clusters. This quantitative method, by design, groups each fund to its closest cohort and captures style-shifting by funds if it occurs.

2. Measuring Fund Abnormal Performance

The abnormal performance of a hedge fund is measured relative to certain benchmarks. Given the wide use of derivatives and dynamic trading strategies among hedge funds, the standard CAPM model cannot adequately capture the risk-return tradeoff for hedge funds. Therefore, we will consider a few alternative choices as performance benchmarks. We plan to use the Fung and Hsieh (FH) 7-factor model (Fung and Hsieh, 2001)⁸ that includes an equity market factor, a size spread factor, a bond market factor, a credit spread factor, and trend-following factors for bond, currency, and commodities. We will also use a modified appraisal ratio of Treynor and Black (1973), calculated by dividing the mean of the monthly abnormal returns by their standard deviation. Finally, we plan to calculate the monthly Sharpe ratio to capture the risk-return tradeoff of hedge fund performance. We will consider the smoothing-adjusted Sharpe ratio to control for illiquidity and smoothing in hedge fund returns, following Getmansky, Lo, and Makarov (2004).

3. Properties of the Strategy Distinctiveness Index

⁸ <http://faculty.fuqua.duke.edu/~dah7/DataLibrary/TF-FAC.xls>

After constructing the *SDI*, based on the cluster styles, we plan to first investigate its properties in several dimensions. Specifically, we will examine the cross-section heterogeneity, persistence, and determinant of the *SDI*. If the *SDI* indeed measures managerial innovation skills, it should have properties that are consistent with its potential of being an effective proxy. For example, if the deviation in hedge fund returns from its peers is driven by innovations in trading strategies and managerial skills, funds should display rather persistent *SDI* over time. In addition, The *SDI*, if a valid proxy for skills, should be correlated with other fund attributes that have been shown to be related to managerial skills. In considering these other attributes, we plan to include fund return volatility (*Vol*), lengths of redemption notice and lockup periods, an indicator variable for personal capital commitment, an indicator variable for high water mark, management fees, incentive fees, fund age, natural logarithm of *AUM*, flow into funds, minimum investment, an indicator variable for using leverage, average monthly net fee returns, FH 7-factor alpha and the corresponding appraisal ratio, and the Sharpe ratio.

4. Strategy Distinctiveness Index and Fund Abnormal Performance

To probe whether *SDI* indeed contains valuable information that could be used to predict future fund abnormal performance, we will adopt both a portfolio sorting and a multivariate predictive regression approach.

Following the portfolio sorting approach, at the beginning of each quarter, we will sort all hedge funds into five portfolios according to their *SDI* levels measured over a previous 24-month period. For each quintile portfolio, we will compute the equally and value weighted average buy-and-hold performance for the subsequent quarter, and then compare the time series average of performance measures between the high and low *SDI* quintile portfolios. We will also consider the performance of these quintile portfolios held for the subsequent 6 months and 1–3 years.

The second approach, the multivariate predictive regression analysis, will enable us to control for other hedge fund characteristics that are known to affect future abnormal performance. To investigate whether the *SDI* has a predictive power for future fund performance after controlling for other fund-specific characteristics, we will estimate the following regression:

$$AbnormalPerformance_{i,t} = c_{0i} + c_{1i}SDI_{i,t-1} + c_{2i}Control_{i,t-1} + e_{i,t} \quad (2)$$

where $AbnormalPerformance_{i,t}$ is the risk-adjusted fund performance within 1 year after the *SDI* is calculated. Specifically, we will consider the alpha, the corresponding appraisal ratio, and the smoothing-adjusted Sharpe ratio.

We will use the lagged control variables to mitigate potential endogeneity problems. The $Controls_{i,t-1}$ will consist of performance volatility measured by the volatility of prior 24-month fund returns in percent (*Vol*), redemption notice period measured in a unit of 30 days, lockup months, indicator variables for whether personal capital is committed and whether there is a high water mark requirement, management fees, incentive fees, ages of funds in years, natural logarithm of *AUM*, flows into funds within the last year as a fraction of *AUM* in percent, average monthly net fee returns in the preceding 24-month period, minimum investment, and an indicator variable for use of leverage. These variables are suggested by the existing literature on hedge fund characteristics and performance. If the distinctiveness index indeed reflects innovative and skillful managerial talents, we should expect its estimated coefficient to be significantly positive.

Our data are a time series and cross-sectional unbalanced panel data. Given the stale price issue for hedge fund data documented by Getmansky, Lo, and Makarov (2004), the resulting alphas may be correlated over time for a specific fund; hence, we must correct for the fund-clustering effect. Moreover, hedge fund performance may also be correlated across funds at a given point of time. Therefore, we need to correct for the time effect. As Petersen (2005) shows, clustering standard error is the preferred approach in addressing the fund effect, while Fama-MacBeth is appropriate for correcting

for the time effect. When both effects exist, we need to address one parametrically and then estimate standard errors clustered on the other dimension. We thus will adopt two approaches. The first approach is the panel regression adjusting for both fund-clustering and time- and style-fixed effects. The second approach is the Fama-MacBeth cross-sectional analysis with style dummies and the Newey-West heteroscedasticity and autocorrelation adjustment (HAC). Since there are only 13 years in our sample, the annual regression, especially for the Fama-MacBeth analysis, will be subject to the issue of limited statistical power. Therefore, our regressions will use data of quarterly frequency.

5. Robustness Tests

Besides the main tests on whether better skilled funds use more distinctive strategy, and deliver superior performance, we also plan to conduct several robustness checks to our hypothesis. First, we will investigate whether the performance predictability of the *SDI* is driven by the hedging hypothesis discussed in Titman and Tiu (2008). Titman and Tiu (2008) show that more skilled hedge fund managers will choose less exposure to systematic risk, hence their funds will exhibit lower R-square of returns on FH 7 factors. It is possible that the low R-square funds also tend to have high *SDI*. Therefore, we will investigate whether the forecasting power of the *SDI* is a manifestation of the hedging effect. Second, we will examine whether our results are due to a dropout bias, resulting from the fact that no performance records are available after funds stop reporting to the TASS database. Third, we will consider alternative measures for strategy distinctiveness, including the absolute correlation-based *SDI* and the TASS style-based *SDI*. Finally, we will use an alternative approach to screen out back filled data.

IV. Project Management: Time Scale and Deliverable Outputs

We plan to complete the empirical analyses and have a polished draft in a year. By the end of the investigation, we will have a paper ready for submission to top finance journals. We would also be more than happy to present our work at Q-Group.

V. CVs of Researchers

Please see attachment at the end.

REFERENCES

- Abraham, J., Goetzmann, W., and Wachter, S., 1994, Homogeneous Groupings of Metropolitan Housing Markets, *Journal of Housing Economics* 3, 186–206.
- Abreu, D., and Brunnermeier, M., 2002, Synchronization Risk and Delayed Arbitrage, *Journal of Financial Economics* 66, 341–360.
- Ackermann, C., McEnally, R., and Ravenscraft, D., 1999, The Performance of Hedge Funds: Risk, Return, and Incentives, *Journal of Finance* 54, 833–874.
- Agarwal, V., Daniel, N., and Naik, N., 2007, Role of Managerial Incentives and Discretion in Hedge Fund Performance, forthcoming in *Journal of Finance*.
- Agarwal, V., and Naik, N., 2000, Multi-Period Performance Persistence Analysis of Hedge Funds, *Journal of Financial and Quantitative Analysis* 35, 327–342.
- Agarwal, V., and Naik, N., 2004, Risks and Portfolio Decisions Involving Hedge Funds, *Review of Financial Studies* 17, 63–98.
- Aggarwal, R., and Jorion, P., 2009, The Risks of Emerging Hedge Fund Managers, *Journal of Investing*.
- Aragon, G., 2007, Share Restriction and Asset Pricing: Evidence from the Hedge Fund Industry, *Journal of Financial Economics* 83, 33–58.
- Berk, Jonathan B., and Green, R. C., 2004, Mutual Fund Flows and Performance in Rational Markets, *Journal of Political Economy* 112(6), 1269–1295.
- Brown, S., and Goetzmann, W., 1997, Mutual Fund Styles, *Journal of Financial Economics* 43, 373–399.
- Brown, S., and Goetzmann, W., 2003, Hedge Funds with Style, *Journal of Portfolio Management* 29, 101–112.
- Brown, S., Goetzmann, W., and Ibbotson, R., 1999, Off-Shore Hedge Funds: Survival and Performance, *Journal of Business* 72, 91–117.
- Brown, S., Goetzmann, W., Liang, B., and Schwarz, C., 2007, Mandatory Disclosure and Operational Risk: Evidence from Hedge Fund Registration, *Journal of Finance*, forthcoming.
- Brown, S., Goetzmann, W., and Ross, S., 1995, Survival, *Journal of Finance* 50, 853–873.
- Brunnermeier, M. and Nagel, S., 2004, Hedge Funds and the Technology Bubble, *Journal of Finance* 59, 2013–2040.
- Carhart, M., 1997, On Persistence in Mutual Fund Performance, *Journal of Finance* 52, 57–82.
- Chen, J., Hong, H., Huang, M., and Kubik, J., 2004, Does Fund Size Erode Mutual Fund Performance? The Role of Liquidity and Organization, *The American Economic Review*

90(5), 1276–1302.

- Cohen, R., Coval, J. D., and Pástor, L., 2005, Judging Fund Managers by the Company That They Keep, *Journal of Finance* 60, 1057–1096.
- Cremers M., and Petajisto A., 2007, How Active is Your Fund Manager? A New Measure That Predicts Performance, forthcoming in *Review of Financial Studies*.
- Ding, B., Getmansky, M., Liang, B., and Wermers, R., 2008, Investor Flows and Share Restrictions in the Hedge Fund Industry, Working Paper.
- Fung, W., and Hsieh, D., 1997, Empirical Characteristics of Dynamic Trading Strategies: The Case of Hedge Funds, *Review of Financial Studies* 10, 275–302.
- Fung, W., and Hsieh, D., 2000, Performance Characteristics of Hedge Funds and CTA Funds: Natural Versus Spurious Biases, *Journal of Financial and Quantitative Analysis* 35, 291–307.
- Fung, W., and Hsieh, D., 2001, The Risk in Hedge Fund Strategies: Theory and Evidence from Trend Followers, *Review of Financial Studies* 14, 313–341.
- Fung, W., and Hsieh, D., 2002, Benchmarks of Hedge Fund Performance: Information Content and Measurement Biases, *Financial Analysts Journal* 58, 22–34.
- Fung, W., Hsieh, D., Naik, N., and Ramadorai, T., 2008, Hedge Funds: Performance, Risk, and Capital Formation, *Journal of Finance*, forthcoming.
- Gray, W., 2009, Do Hedge Fund Managers Identify and Share Profitable Ideas?, University of Chicago Booth School of Business Working Paper.
- Getmansky, M., Lo, A., and Makarov, I., 2004, An Econometric Model of Serial Correlation and Illiquidity in Hedge Fund Returns, *Journal of Financial Economics* 74, 529–609.
- Goetzmann, W., Ingersoll, J., and Ross, S., 2003, High-Water Marks and Hedge Fund Management Contracts, *Journal of Finance* 58, 1685–1717.
- Goetzmann, W., and Wachter, S., 1995, Clustering Methods for Real Estate Portfolios, *Real Estate Economics* 23, 271–310.
- Griffin, J., and Xu, J., 2007, How Smart are the Smart Guys? A Unique View from Hedge Fund Stock Holdings, *Review of Financial Studies*, forthcoming.
- Hunter, D., Kandel, E., Kandel, S., and Wermers, R., 2009, Endogenous Benchmarks, University of Maryland Working Paper.
- Ibbotson, R., and Chen, P., 2006, The A,B,Cs of Hedge Funds: Alphas, Betas, and Costs, Yale ICF Working Paper No. 06–10.
- Jagannathan R., Malakhov, A., and Novikov, D., 2006, Do Hot Hands Exist Among Hedge Fund Managers? An Empirical Evaluation, NBER Working Paper Series, Vol. w12015.

- Kacperczyk, M., and Seru, A., 2007, Fund Manager User of Public Information: New Evidence on Managerial Skills, *Journal of Finance* 62, 485–528.
- Kacperczyk, M., Sialm, C., and Zheng, L., 2005. On the Industry Concentration of Actively Managed Equity Mutual Funds, *Journal of Finance* 60, 1983–2011.
- Kacperczyk, M., Sialm, C., and Zheng, L., 2007. Unobserved Actions of Mutual Funds, *Review of Financial Studies*, forthcoming.
- Kosowski, R., Naik, N. Y., and Teo, M., 2007, Do Hedge Funds Deliver Alpha? A Bayesian and Bootstrap Analysis, *Journal of Financial Economics* 84, 229–264.
- Li, H., Zhang, X., and Zhao, R., 2007, Investing in Talents: Manager Characteristics and Hedge Fund Performances, Working Paper, University of Michigan.
- Liang, B., 1999, On the Performance of Hedge Funds, *Financial Analysts Journal* 55, 72–85.
- Liang, B., 2000, Hedge Funds: The Living and the Dead, *Journal of Financial and Quantitative Analysis* 35, 309–326.
- Liang, B., and Park, H., 2008, Share Restriction, Liquidity Premium, and Offshore Hedge Funds, Working Paper, University of Massachusetts.
- Lo, A., 2002, The Statistics of Sharpe Ratios, *Financial Analysts Journal* 58, 36–52.
- Malkiel, B., and Saha, A., 2005, Hedge Fund: Risk and Return, *Financial Analysts Journal* 61, No. 6: 80-88.
- Petersen, M., 2005, Estimating Standard Errors in Finance Panel Data Sets: Comparing Approaches, Working Paper, Northwestern University.
- Stein, J., 2008, Conversations among competitors, *American Economic Review*, forthcoming.
- Stein, J., 2009, Presidential Address: Sophisticated Investors and Market Efficiency, *Journal of Finance* 64, 1517-1548
- Titman, S., and Tiu, C., 2008, Do the Best Hedge Funds Hedge? Working Paper, University of Texas at Austin.
- Treynor, J., and Black, F., 1973, How to Use Security Analysis to Improve Portfolio Selection, *Journal of Business* 46, 66–86.

Curriculum Vitae

ZHENG SUN

The Paul Merage School of Business, Office 437
University of California - Irvine
Irvine, CA 92697

Email: zsun@merage.uci.edu
Tel: 949-824-6907

ACADEMIC APPOINTMENT

University of California, Irvine, CA **IRVINE, CA**
Assistant Professor, The Paul Merage School of Business, since July 2007

EDUCATION

Stern School of Business, New York University	NEW YORK, NY
<i>Ph.D. in Finance, May 2007</i>	
Department of Economics, Ohio State University	COLUMBUS, OH
<i>M.A. in Economics, August 2001</i>	
Department of Finance, Peking University	BEIJING, CHINA
<i>B.A. in Economics, August 2000</i>	

RESEARCH

AREAS OF INTEREST

Investments, Banking, Empirical Market Microstructure, Financial Econometrics

PUBLICATIONS

Institutional Demand Pressure and Cost of Corporate Loans (with Victoria Ivashina), Journal of Financial Economics, Forthcoming

http://papers.ssrn.com/sol3/papers.cfm?abstract_id=1299300

Institutional Stock Trading on Loan Market Information (with Victoria Ivashina), Journal of Financial Economics, Forthcoming

http://papers.ssrn.com/sol3/papers.cfm?abstract_id=972044

WORKING PAPERS

When is Noise Not Noise – A Microstructure Estimate of Realized Volatility (with Robert Engle), 2006

Link to paper: http://papers.ssrn.com/sol3/papers.cfm?abstract_id=1293630

Clustered Institutional Holdings and Stock Co-Movement, 2007

http://papers.ssrn.com/sol3/papers.cfm?abstract_id=1332201

The Road Less Traveled: Strategy Distinctiveness and Hedge Fund Performance (with Ashley Wang and Lu Zheng), 2009

http://papers.ssrn.com/sol3/papers.cfm?abstract_id=1337424

Do Active Funds Perform Better in Down Markets – New Evidence from Cross-Sectional Study? (with Ashley Wang and Lu Zheng), 2009

http://papers.ssrn.com/sol3/papers.cfm?abstract_id=1474083

WORK IN PROGRESS

Performance of Socially Connected Money Managers (with Ashley Wang and Lu Zheng)

Hedge funds' use of credit lines (with Jiekun Huang and Victoria Ivashina)

TEACHING

Winter 2009, 2010	Derivatives FTMBA and Undergrad elective course
Winter 2008, Fall 2009, 2010	Introduction to Managerial Finance, UCI Undergrad core course
Fall 2007, 2008	Introduction to Managerial Finance, UCI FTMBA core course, FEMBA core course
Fall 2006, Fall 2005	Foundations of Financial Markets Core Enhancement, NYU Full responsibility for the core undergraduate course
Summer 2005	Foundations of Financial Markets, NYU Full responsibility for the core undergraduate course Student Excellence in Teaching Award
Summer 2004 – Spring 2006	Teaching Assistant - Debt Instruments and Markets, NYU MBA course (Prof. Jacob Boudoukh, Prof. Matthew Richardson, Prof. Jennifer Carpenter)
Spring 2004	Teaching Assistant - Financial Theory II , NYU Ph.D. course (Prof. Robert Engle)

AWARDS AND FELLOWSHIPS

- Finalist for the 2006 Lehman Brothers Fellowship for Research Excellence in Finance
- New York University Alumni Fellowship, Stern School of Business, New York University 2006-2007
- Doctoral Fellowship, Stern School of Business, New York University 2001-2006
- Student Excellence in Teaching Award, Stern School of Business, Summer 2005
- University Fellowship, The Ohio State University, 2000-2001

SCHOLARLY ACTIVITIES

INVITED PRESENTATIONS

- 2010 China International Conference in Finance, July 2010, Beijing
- American Finance Association Annual Conference, January 2010(co-author presented), Atlanta
- UCI Assistant Professor Colloquia Series, November 2009, Irvine
- UCI Paul Merage School of Business Brownbag seminar, June 2009, Irvine
- Financial Intermediation Research Society 2009 meetings (co-author presented), May 2009, Prague
- UCI Assistant Professor Colloquia Series, February 2009, Irvine
- American Finance Association Annual Conference, (co-author presented), January 2008, New Orleans
- European Finance Association Annual Conference (co-author presented), August 2007, Slovenia
- NBER Summer Institute, Corporate Finance Meeting (co-author presented), August 2007, Boston
- The Ohio State University, February 2007
- University of Texas, Austin, January 2007
- University of California, Irvine, January 2007
- Massachusetts Institute of Technology, January 2007
- Lehman Brothers, Inc, December 2006, New York
- FMA Annual Meeting, October 2006, Salt Lake City
- London Business School Transatlantic Conference, May 2005, London

- European Finance Association Annual Conference, August 2005, Moscow
- NBER Market Microstructure Working Group Meeting, May 2005, Boston
- NYU Solomon Center Financial Econometric Group Seminar, April 2005, New York
- Realized Volatility Conference (Poster Session), November 2003, Montreal

DISCUSSIONS

- European Finance Association Annual Conference, August 2005, Moscow
- 2009 FMA Annual Meeting, Reno Nevada
- 2010 China International Conference in Finance, July 2010, Beijing

SESSION CHAIR

- 2009 FMA Annual Meeting, Reno Nevada

REFEREE

- Journal of Finance
- Journal of Empirical Finance
- Journal of Quantitative Finance

PUBLIC MEDIA

“Institutional Stock Trading on Loan Market Information” (with Victoria Ivashina) cited by The Wall Street Journal, March 13, 2008 and July 3, 2010

SCHOOL AND UNIVERSITY SERVICE

Colloquium Series Organizer, The Paul Merage School of Business

Finance Colloquium, The Paul Merage School of Business, UCI, 2007-2008

Committee Service, The Paul Merage School of Business

Undergraduate Committee, The Paul Merage School of Business, UCI, 2007-2008

Curriculum Innovation Committee, The Paul Merage School of Business, UCI, 2008-2009

PhD Committee, The Paul Merage School of Business, UCI, 2009-2010

ADVISING

Member, Dissertation Committee

Yong Rin Park, The Paul Merage School of Business, June 2009

Zachary Nye, The Paul Merage School of Business, August 2009

AFFILIATIONS

- American Finance Association
- Western Finance Association
- European Finance Association
- Financial Management Association

Curriculum Vitae

Ashley W. Wang

Capital Markets Group
Research and Statistics Division
Board of the Governors
Washington, DC20551

Office: 202-452-3122
Email: ashley.wang@frb.gov

EDUCATION

2003 PhD, Finance
1998 MS
1996 BS

Anderson School, UCLA, Los Angeles, CA
Johns Hopkins University
Peking University

PROFESSIONAL APPOINTMENTS

2010-present	Economist	Federal Reserve Board, Washington DC
2003-2010	Assistant Professor of Finance	Paul Merage School of Business, UC Irvine
1998-2003	Research and Teaching Assistant	Finance Dept. Anderson School at UCLA
1996-1998	Research Assistant	Johns Hopkins University

RESEARCH INTERESTS

Empirical Asset Pricing, Liquidity, Institutional Investment Behavior, Mutual Funds and Hedge Funds

GRANTS, AWARDS AND HONORS

Council on Research, Computing & Library Resources Award, 2006-2009
Elsevier ADVANCE Travel Award: 2008, 2009
Research Grant by the Don Beall Center for Innovations and Entrepreneurship, 2008
Faculty Career Development Award, 2006-2007
Allstate Dissertation Fellowship, UCLA, 2002
AFA Doctoral Student Traveling Grant, 2002
UC Regents Fellowship, 1998-2001
Summer Doctoral Fellowship, UCLA, 1998-2001
Fellowship, the Johns Hopkins University, 1996-1998

PUBLISHED PAPERS

1. "Estimation and Test of A Simple Model of Inter-temporal Capital Asset Pricing," with Michael Brennan and Yihong Xia, 2004, Journal of Finance 59, 1743-1776.
2. "Electricity Forward Prices: A High Frequency Empirical Analysis," with Francis Longstaff, 2004, Journal of Finance 59, 1877-1900
3. "Institutional Ownership and Credit Spreads: An Information Asymmetry Perspective" with Gaiyan Zhang, 2009, Journal of Empirical Finance 16, 587-612..
4. "Mispricing Return Premium" with Michael Brennan, forthcoming at Review of Financial Studies, 2010.

WORKING PAPERS

5. "Road Less Travelled: Strategy Distinctiveness and Hedge Fund Performance", with Zheng Sun and Lu Zheng, 2010, invited submission by Review of Financial Studies.
6. "Aggregate Hedge Fund Flows and Asset Returns", with Chris Schwarz and Lu Zheng, 2009
7. "Do Active Funds Perform Better In Down Markets?- New Evidence from Cross-Sectional Study", with Zheng Sun and Lu Zheng, 2009.
8. "Systematic Mispricing", with Michael Brennan, 2009
9. "Institutional Equity Flows, Liquidity Risk and Asset Pricing", 2007.

PROFESSIONAL ACTIVITIES

Presentations at National Conferences and Finance Workshops:

University of Hawaii, May 2010
Temple University, Feb. 2010
Rutgers University, Feb. 2010
Federal Reserve Board at Washington DC, Feb. 2010
University of Texas, Dallas, Jan. 2010
AFA at Atlanta, Jan. 2010
University of Maryland, College Park, Sep. 2009
Arizona State University, Sep. 2009
George Washington University, Sep. 2009
Cheung Kong GSB Summer Research Conferences at Guilin, 2009
Third Singapore International Conference on Finance, 2009
WFA at San Diego, 2009
Financial Intermediation Research Society Conference in Prague, 2009
University of California, Riverside, 2009
Santa Clara University, 2009
Cal State University at Fullerton, 2008

WFA at Big Sky, Montana, 2007
BGI Asset Pricing Workshop, 2007
17 CFEA Conference in Atlanta, 2006
University of Georgia, Athens, 2006
University of Vienna, 2006
Hong Kong University of Science and Technology, 2006
University of Southern California, 2006
California State University at Fullerton, 2006
China International Conference in Finance, 2004, 2006
Western Finance Association Annual Conference at Vancouver, 2004
UC Davis, Graduate School of Management Colloquium, 2003, 2004, 2007
Rice University, Finance Seminar, 2003
UC Riverside, Finance Seminar, 2003
UC Irvine, Finance Seminar, Jan. 2003, and Feb. 2003
Washington University at St. Louis, Finance Seminar, 2003
University of Utah Finance Seminar, 2003
UCLA, Finance Seminar, 2002
MSRI Event Risk Conference at New York, 2002
The University of California Energy Institute Friday Seminar, 2002
CIBER Conference in Finance 2000, UCLA

Conferences

Western Finance Association annual conferences, 2000 – 2010
American Finance Association annual conferences, 2002 - 2010
China International Conference for Finance, 2004, 2006
UC Davis Annual Finance and Accounting Conference at Napa, 2004-2007
NBER Asset Pricing Program Meeting, Mar. 2001, Mar. 2002
Univ. of Michigan Mitsui Life Symposium: “Institutional Investors: Issues in Asset Management and Governance”, Jun. 2005
Finance Research Association annual conference, 2005-2006

Referee:

Energy Journal
International Review of Finance
Journal of Banking and Finance
Journal of Empirical Finance
Journal of Finance
Journal of Financial and Quantitative Analysis
Journal of Financial Markets

Professional Affiliation:

American Finance Association, member
Western Finance Association, member
Society of Financial Studies, member

Curriculum Vitae

Lu Zheng

Office Address

The Paul Merage School of Business
University of California, Irvine
Irvine, CA 92697-3125
Office Phone (949) 824-8365
Office Fax (949) 725-2870
Email: luzheng@uci.edu

Education

Ph.D. in Finance, Yale University, May 1999.
B.A., Agnes Scott College, Atlanta, 1993.

Academic Appointments

Associate Professor of Finance, The Paul Merage School of Business, University of California, Irvine, 2006 - present.
Assistant Professor of Finance, The Stephen M. Ross School of Business, University of Michigan, 1999 - 2006, approved for promotion to associate professor with tenure in May 2006.
Lecturer in Finance, University of Michigan Business School, 1998 - 1999.

Publications

- "Side-by-Side Management of Hedge Funds and Mutual Funds," 2010, with Tom Nohel and Zhi Wang, *The Review of Financial Studies*, 23(6), 2342-2373.
- "Unobserved Actions of Mutual Funds," 2008, with Marcin Kacperczyk and Clemens Sialm, *The Review of Financial Studies*, 21, 2379-2416. (Lead article)
- "Investor Flows and Stock Market Returns," 2009, with Brian Boyer, *Journal of Empirical Finance*, 16 (1), 87-100.
- "The ABCs of Mutual Funds: On the Introduction of Multiple Share Classes," 2009, with Vikram Nanda and Jay Wang, *Journal of Financial Intermediation*, 18 (3), 329-361.
- "Tax Loss Selling and the January Effect: Evidence from Municipal Bond Closed-end Funds," 2006, with Laura Starks and Li Yong, *The Journal of Finance*, 61 (6), 3049-3067.
- "Industry Concentration and Mutual Fund Performance," 2007, with Marcin Kacperczyk and Clemens Sialm, *The Journal of Investment Management*, 5 (1), 50-64.
- "Are Investors Moonstruck? – Lunar Phases and Stock Returns," 2006, with Kathy Yuan and Qiaoqiao Zhu, *Journal of Empirical Finance* 13(1), 1-23.
- "Out of Sight, Out of Mind: the Effects of Expenses on Mutual Fund Flows," 2005, with Brad Barber and Terrance Odean, *The Journal of Business* 78 (6), 2095-2120.
- "On the Industry Concentration of Actively Managed Mutual Funds," 2005, with Marcin T. Kacperczyk and Clemens Sialm, *The Journal of Finance* 60 (4), 1983-2011.

“Family Values and the Star Phenomenon: Strategies of Mutual Fund Families,” 2004, with Vikram Nanda and Jay Wang, *The Review of Financial Studies* 17 (3), 667-698.

“Institutional Trading and Stock Returns,” 2004, with Fang Cai, *Finance Research Letters* 1 (3), 178-189.

“Is Money Smart? – A Study of Mutual Fund Investors’ Fund Selection Ability,” 1999, *The Journal of Finance* 54 (3), 901 – 933.

Invited Book Chapters

“The Behavior of Mutual Fund Investors,” Handbook of Financial Intermediation and Banking, Arnoud Boot and Anjan Thakor, eds., North Holland/Elsevier Handbook of Readings.

Working Papers

“The Frequency of Mutual Fund Portfolio Disclosure,” with Weili Ge.

“The Road Less Traveled: Strategy Distinctiveness and Hedge Fund Performance,” 2008, with Zheng Sun and Ashley Wang.

“Aggregate Hedge Flows and Asset Returns,” 2008, with Ashley Wang.

“The Good, the Bad or the Expensive? Which Mutual Fund Managers Join Hedge Funds?” 2009, with Prachi Deuskar, Joshua Matthew Pollet and Zhi Jay Wang.

“Does Your Fund Manager Trade on the News? Media Coverage, Mutual Fund Trading and Performance,” 2009, with Lily H. Fang and Joel Peress

Research in Progress

“Investor Expectations and Stock Market Movements,” with Akiko Kamesaka and Xiaoyan Yu.

Press Appearances

Money Management Executive, March 19, 2007.

Institutional Investor Magazine, February 7, 2007.

Registered Rep, February 1, 2007.

Bloomberg.com, November 27, 2006.

Wall Street Journal, November 1, 2006.

The Wall Street Journal Asia, October 16, 2006.

Wall Street Journal, October 14, 2006.

Wall Street Journal, March 22, 2006.

Business Week, March 20, 2006.

New York Times, January 8, 2006.

The Associated Press, August 2, 2005.

CBS Market Watch, July 31, 2005.

Investor’s Business Daily, February 10, 2005.

The Jerusalem Post, November 19, 2004.

Ann Arbor News, May 24, 2004.

The Evening Standard, May 18, 2004.

Michigan Radio, May 10, 2004.

Chicago Sun-Times, May 10, 2004.

Christian Science Monitor, May 10, 2004.

Ann Arbor News, May 6, 2004.
TheStreet.com, April 26, 2004.
Financial Times, April 19, 2004.
New York Times, April 11, 2004.
Wall Street Journal, March 31, 2004.
Institutional Investor, March 1, 2004.
The Sunday Oregonian, December 21, 2003.
Financial Planning Magazine, November 26, 2003.
Investors' Business Daily, November 25, 2003.
Wall Street Journal, November 25, 2003.
Treasure Coast Business Journal, May 1, 2003.
NPR, Michigan Radio, April 21, 2003.
Chicago Sun-Times, March 17, 2003.
Wall Street Journal, March 14, 2003.
NPR, Market Place, March 19, 2002.
Financial Times, January 10, 2002.
Economist, October 20th, 2001.
Business Week, October 23, 2000.
MBA Bullet Point, April 25 – May 8, 2000.
TheStreet.com, June 19, 1999.
Money Magazine, May, 1998.

Editorial and Referee Duties for Professional Journals

Associate Editor, Review of Financial Studies, 2009-present.

Referee for *American Economic Review*, *Chinese Economic Review*, *European Financial Management*, *Financial Management*, *Finance Research Letters*, *Journal of Banking and Finance*, *Journal of Business*, *Journal of Empirical Finance*, *Journal of Finance*, *Journal of Financial Economics*, *Journal of Financial and Quantitative Analysis*, *Journal of Financial Markets*, *Journal of Financial Research*, *Journal of International Business Studies*, *Journal of The Japanese and International Economies*, *The National Science Foundation*, *Review of Finance*, *Review of Financial Studies*, *Real Estate Economics*, *Pacific-Basin Finance Journal*, and *Research Grants Council of Hong Kong*.

Papers Presented at Professional Meetings

American Finance Association Meetings, Atlanta, 2010
 Western Finance Association Meeting, Hawaii, June 2008.
 Inquire Europe and Inquire U.K. Seminar in Brighton (U.K.), March 2007.
 Midwest Finance Association Annual Meetings, February 2007.
 FIRS Conference on Banking, Corporate Finance and Intermediation, Shanghai, May 2006.
 Utah Winter Finance Conference, Utah, February 2006.
 Financial Research Association Meeting, Las Vegas, December 2005.
 Conference on Financial Economics and Accounting at the University of North Carolina, November 2005.
 European Finance Association Annual Meetings, Moscow, August 2005.
 American Finance Association Annual Meetings, Philadelphia, January 2005.
 Conference on Delegated Portfolio Management, The University of Oregon and The Journal of Financial Economics, September 2004.
 6TH Annual Tax Symposium, University of North Carolina at Chapel Hill, March 2003.
 American Finance Association Annual Meetings, Washington D.C., January 2003.
 CIRANO mutual fund conference, Montreal, December 2002.
 Conference on Distribution and Pricing of Delegated Portfolio Management, Wharton Financial Institutions Center, May, 2002.
 Financial Management Association Meeting, Toronto, October, 2001.
 European Finance Association Meeting, Barcelona, August 2001.
 Western Finance Association Meeting, Tucson, June 2001.
 Tel-Aviv Conference in Accounting and Finance, July, 2001.

NBER Behavioral Finance Conference, November 2000.
Conference on Financial Economics and Accounting, UT Austin, November 1999.
American Finance Association Annual Meeting, Chicago, January 1998.
European Finance Association Annual Meeting, Vienna, August 1997.
Financial Management Association International Annual Meeting, Zurich, May 1997.
Eastern Finance Association Annual Meeting, Panama City, April 1997.
Conference on Financial Economics and Accounting, Rutgers, November 1996.
New England Doctoral Students Symposium, Amherst, September 1996.

Other Recent Presentations

University of California, Riverside, March, 2010
University of Oregon, November, 2008.
Georgetown University, October 2008.
Hedge Fund Symposium at Loyola University, March 2007.
U.S. Securities and Exchange Commission, 2006.
Investment Company Institute, 2006.
The Center for Investment and Wealth Management, UC Irvine, 2006.
Yale University School of Management 2006.
University of California, Irvine, 2006.
Hong Kong University of Science and Technology, 2005.
University of Binghamton, March 2005.
University of Florida, February 2004.
Hong Kong University of Science and Technology, May 2002.
University of Texas at Austin, March 2002.
University of Illinois at Urbana-Champaign, December 2001.
Michigan State University, October 2001.
Hosmer Lunch Seminar, University of Michigan Business School, December 2001 and April 2000.

Work Experience

Internship, Federal Reserve Bank of Atlanta, 1993.

Honors

Research Grant, The Don Beall Center for Innovation and Entrepreneurship, 2008.
“Best Paper in Financial Institutions” at the 2007 Midwest Finance Association Conference.
Inquire (the Institute for Quantitative Investment Research) UK, Research Grant, 2006.
Nomination for the Smith Breeden prize for the best paper in The Journal of Finance, 2005.
The 2005 NTT Fellowship.
Inquire (the Institute for Quantitative Investment Research) Research Grant, 2004.
Q-group Research Grant, 2000.
Nomination for the Smith Breeden prize for the best paper in The Journal of Finance, 1999.
Doctoral Fellowship, Yale University.
Phi Beta Kappa National Honor Society.